1. **Task 1 - Python Requests**
   - **a)** Use the requests python package to fetch data from the following API link - https://jsonplaceholder.typicode.com/posts
   - **b)** Fetch all posts and print them
   - **c)** Count the distinct number of users
   - **d)** Which user has the highest number of posts ?
   - **e)** What is average word length of post title ?
   - **f)** You may use pandas, numpy and any other packages for the above requirement

2. **Task 2 - html parser**
   - a) Read the following html text in python as a string

```
<html><head><title>The Dormouse's story</title></head><body>
<p class="title">&lt;b&gt;The Dormouse's story&lt;/b&gt;</p>
<p class="story">Once upon a time there were three little sisters; and their
names were
<a href="http://example.com/elsie" class="sister" id="link1">Elsie</a>,
<a href="http://example.com/lacie" class="sister" id="link2">Lacie</a> and
<a href="http://example.com/tillie" class="sister" id="link3">Tillie</a>;
and they lived at the bottom of a well.</p>
<p class="story">...</p>
```

   - b) Use BeautifulSoup package to print the following
     - i. All href links in the given html text
     - ii. All class values for <a> tags
     - iii. Extra points for also writing a recursive python code which is capable to parse the above html text and give the output of point (i) and (ii) without using BeautifulSoup package

3. **Task 3 - FastAPI -** Write a python code to create the following API using fastAPI
   - a) To accept 2 numbers and return their sum. Ensure the API is able to give proper error value when string is entered in place of numbers
   - b) To accept lower case string and return the string in all capital letters

4. **Task 4 - Data analysis -** Download the following data and analyse using python - https://maven-datasets.s3.amazonaws.com/Global+Electronics+Retailer/Global+Electronics+Retailer.zip

5. **Task 5 - SQL topper in class -** Write a sql code to identify the following
   - a) The second topper in class

Name	Marks
Sahil	95
Kaushik	90
John	89
Kara	87
Simpson	97

   - b) The second topper in class. If 2 candidates have same marks then the one with their name first in alphabetical order is given the better(lower) rank

Name	Marks
Sahil	95
Kaushik	97
John	89

Kara	87
Simpson	97

c) The second topper(s) in class. If multiple candidates have same marks, they are given the same rank, so multiple individuals can have rank 1.

Name	Marks
Sahil	95
Kaushik	97
John	95
Kara	87
Simpson	97

6. **Task 6 - Documentation -** Upload all the projects on github with proper documentation and details and share the link of the repository with us.